

# Internship Report

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## **Abstract**

This internship focused on understanding Quantum Speed Limits (QSLs), which define the fundamental limits on the time required for a quantum system to evolve from one state to another. The project explored various formulations of QSLs, including the Mandelstam-Tamm bound and the Margolus-Levitin bound, and their applications in two state quantum systems.

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# Chapter 1

## Introduction

This chapter introduces the background of the internship, the motivation behind the project, and the main objectives that were set out to be achieved.

### 1.1 Background

Describe the context of your work at SRM AP.

### 1.2 Objectives

To learn about Quantum Speed Limits and apply it to a two state quantum system.

# Chapter 2

## Methodology

This chapter details the methods, tools, and techniques used during the internship.

### 2.1 Tools and Technologies

Mention the software, hardware, or theoretical frameworks utilized.

### 2.2 Implementation Details

Describe how the project was implemented step-by-step.

# Chapter 3

## Application to a Two-State System

This chapter applies the theoretical framework of Mandelstam and Tamm's Quantum Speed Limit discussed in the methodology to a specific problem: a two-state quantum system.

### 3.1 System Hamiltonian and Time Evolution

Consider a two-state quantum system governed by the Hamiltonian:

$$H = \frac{\hbar\omega_0}{2}\sigma_z = \frac{\hbar\omega_0}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (3.1)$$

where  $\sigma_z$  is the Pauli Z matrix. The unitary time evolution operator is given by:

$$U(t) = e^{-iHt/\hbar} = \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{i\omega_0 t/2} \end{pmatrix} \quad (3.2)$$

### 3.2 Energy Variance and the MT Bound

To apply the Mandelstam-Tamm bound, we first compute the energy variance  $\Delta H$  of the initial state. The expectation value of the Hamiltonian is:

$$\langle H \rangle = \langle \psi(0) | H | \psi(0) \rangle = \frac{\hbar\omega_0}{2} \left( \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \right) = \frac{\hbar\omega_0}{2} \cos \theta \quad (3.3)$$

Since  $H^2 = \left(\frac{\hbar\omega_0}{2}\right)^2 I$ , the expectation value of  $H^2$  is simply  $\langle H^2 \rangle = \left(\frac{\hbar\omega_0}{2}\right)^2$ . The energy fluctuation (standard deviation) is therefore:

$$\Delta H = \sqrt{\langle H^2 \rangle - \langle H \rangle^2} = \sqrt{\left(\frac{\hbar\omega_0}{2}\right)^2 - \left(\frac{\hbar\omega_0}{2} \cos \theta\right)^2} = \frac{\hbar\omega_0}{2} \sin \theta \quad (3.4)$$

According to the Mandelstam-Tamm bound, the time  $t$  required for the system to evolve to a state separated by a Bures angle  $\Theta$  from the initial state is bounded by:

$$t \geq \frac{\hbar\Theta}{\Delta H} = \frac{\hbar\Theta}{\frac{\hbar\omega_0}{2}\sin\theta} = \frac{2\Theta}{\omega_0\sin\theta} \quad (3.5)$$

For the system to reach an orthogonal state, the distance is  $\Theta = \pi/2$ . The minimum time required to reach this orthogonal state is thus:

$$t \geq \frac{2 \cdot (\pi/2)}{\omega_0\sin\theta} \implies \tau_{QSL} = \frac{\pi}{\omega_0\sin\theta} \quad (3.6)$$

This indicates that the speed of evolution depends on the initial state parameter  $\theta$ . The fastest evolution to an orthogonal state occurs when  $\theta = \pi/2$  (states on the equator of the Bloch sphere), yielding a minimum time of  $\tau_{QSL} = \pi/\omega_0$ .

### 3.3 Generalized Mandelstam-Tamm Bound

Let the initial state of the system be parameterized by an angle  $\theta$  on the Bloch sphere:

$$|\psi(0)\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle \quad (3.7)$$

The state at a later time  $t$  is:

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle = \cos\frac{\theta}{2}e^{-i\omega_0 t/2}|0\rangle + \sin\frac{\theta}{2}e^{i\omega_0 t/2}|1\rangle \quad (3.8)$$

The inner product between the initial and evolved state is:

$$\langle\psi(0)|\psi(t)\rangle = \cos^2\frac{\theta}{2}e^{-i\omega_0 t/2} + \sin^2\frac{\theta}{2}e^{i\omega_0 t/2} = \cos\left(\frac{\omega_0 t}{2}\right) - i\cos\theta\sin\left(\frac{\omega_0 t}{2}\right) \quad (3.9)$$

The squared magnitude of this overlap gives the fidelity:

$$|\langle\psi(0)|\psi(t)\rangle|^2 = \cos^2\left(\frac{\omega_0 t}{2}\right) + \cos^2\theta\sin^2\left(\frac{\omega_0 t}{2}\right) \quad (3.10)$$

The Bures angle  $\alpha(t)$  between the initial and evolved state is defined as:

$$\alpha(t, \theta) = \arccos(|\langle\psi(0)|\psi(t)\rangle|) = \arccos\sqrt{\cos^2\left(\frac{\omega_0 t}{2}\right) + \cos^2\theta\sin^2\left(\frac{\omega_0 t}{2}\right)} \quad (3.11)$$

### 3.4 Computational Analysis

Using Python, we computationally plotted the Bures angle  $\alpha(t)$  as a function of the dimensionless time  $\omega_0 t$  for various initial states (different values of  $\theta$ ) to analyze the generalized Mandelstam-Tamm (MT) bound.

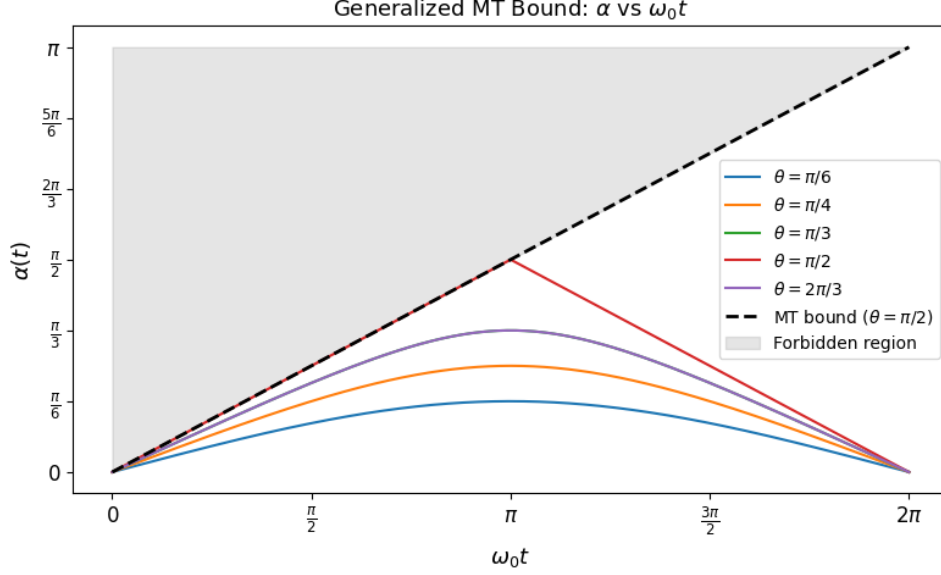


Figure 3.1: Generalized Mandelstam-Tamm Bound:  $\alpha(t)$  vs  $\omega_0 t$  for different initial states  $\theta$ . The black dashed line represents the MT bound for orthogonal states ( $\theta = \pi/2$ ), and the shaded gray region is forbidden by the bound.

As observed in Figure 3.1, for all initial states, the time evolution strictly respects the generalized MT bound. The evolution is fastest for  $\theta = \pi/2$  (corresponding to states on the equator of the Bloch sphere), which reaches an orthogonal state ( $\alpha = \pi/2$ ) at  $\omega_0 t = \pi$ . For other values of  $\theta$ , the system never reaches an orthogonal state, and the curve stays below the theoretical maximum bound.



# Chapter 4

## Results and Conclusion

This chapter presents the results obtained from the experiments or tasks carried out during the internship.

### 4.1 Observations

Detail your findings here. You can include figures and tables.

### 4.2 Discussion

Discuss the significance of your results and how they meet the objectives.

### 4.3 Summary

Briefly restate the main achievements of the project.

### 4.4 Future Work

Mention any extensions or improvements that could be made to the project in the future.

# Bibliography

- [1] John Doe and Jane Smith. An example paper on machine learning. *Journal of Examples*, 2026.